

Evan Pai

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EDUCATION

- **University of British Columbia** Vancouver, BC
Master of Data Science; CAD \$10,000 MDS Scholarship Expected Jun 2027
- **National Tsing Hua University** Hsinchu, Taiwan
Bachelor of Computer Science; GPA: 3.97/4.30 Sep 2021 - Jul 2025

EXPERIENCE

- **National Tsing Hua University, LSA Lab** Hsinchu, Taiwan
Research Assistant, Industry-Academia Collaboration (AI Infrastructure) Jul 2022 - Jul 2025
 - **ML-Based Workload Scheduler:** Built a scheduler that profiles jobs and classifies them with XGBoost on system features, routing each to CPU, NVIDIA, or AMD GPU clusters to maximize utilization.
 - **LLMs on AMD GPUs:** Ported LLM frameworks to ROCm, deployed LLaMA 70B on AMD MI210 GPUs, and benchmarked vLLM, DeepSpeed, and quantization against NVIDIA baselines.
- **IBM** Taipei, Taiwan
Data Scientist Intern Jul 2024 - Aug 2024
 - **Predictive Modeling:** Developed XGBoost and LightGBM models on internal business data, benchmarked against logistic regression and random forest with cross-validation and tuning.
 - **Data Pipelines:** Built Python pipelines with pandas for cleaning and feature engineering, and integrated models into internal applications.
- **National Center for High-performance Computing (NCHC)** Hsinchu, Taiwan
Software Engineer Intern: quantum ML data re-uploading classifier in Python (PennyLane) Jul 2023 - Aug 2023
- **National Tsing Hua University** Hsinchu, Taiwan
Teaching Assistant: Operating Systems, Parallel Programming, HPC Cluster Practice Aug 2023 - Aug 2025
 - **Teaching:** Taught labs for 100+ students per course; designed lab assignments and held TA office hours.

PROJECTS

- **On-Premise RAG System (LLaMA 3.1 70B, vLLM, 4× H100):** Built a local retrieval-augmented generation assistant over internal documents for NTHU's computer center, keeping confidential data off the cloud; tuned vLLM batching and kernels for 3× throughput.

AWARDS AND COMPETITIONS

- **1st Place, 2023 APAC HPC-AI Competition** Sydney, Australia | Feb 2024
1st of 21 teams; accelerated BLOOM LLM inference 11.4× via tensor parallelism, DeepSpeed, and NCCL tuning.
- **2nd Place, ISC 2024 Student Cluster Competition** Hamburg, Germany | May 2024
Sped up the Neko CFD solver 18.5× on 10× NVIDIA H100 under a power cap via profiling and communication tuning.
- **3rd Place, SC24 Student Cluster Competition** Atlanta, US | Nov 2024
Optimized MLPerf Stable Diffusion with TensorRT; trained YOLO11x object detection (3rd); reproduced DataLife.
- **2nd Place, 2023 High Performance Application Competition** Hsinchu, Taiwan | Aug 2023
Ran the FourCastNet AI weather model end to end: multi-node training, inference, and visualization.

PUBLICATION

- C.-A. Pai *et al.*, "Critique of 'Data Flow Lifecycles for Optimizing Workflow Coordination'," **IEEE Transactions on Parallel and Distributed Systems (TPDS)**, 2025. DOI: 10.1109/TPDS.2025.3627428

SKILLS

- **Languages:** Python, C/C++, Bash, SQL
- **Software Engineering:** Linux, object-oriented design, microservices, AWS, CI/CD, Docker, Git/GitHub
- **Machine Learning and Data:** PyTorch, Hugging Face, XGBoost, LightGBM, scikit-learn, pandas, feature engineering, cross-validation, RAG
- **AI Systems and HPC:** CUDA, ROCm, vLLM, DeepSpeed, TensorRT, quantization, NCCL, MPI, OpenMP, LLM inference optimization, profiling